



Digital Order Management with Database System and Retail Analytics

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Database Systems Engineering and Distributed Backend Development (25CS1302E) | KLH Bachupally Campus, CSE | PBL 2026-27

Abstract

- A web-based portal that centralizes retail and e-commerce operations for administrators, staff and customers, with JWT role-based access, order-lifecycle management, inventory alerts, loyalty rewards and a Chart.js analytics dashboard.
- Built with React.js, Node.js/Express.js and MySQL (Sequelize ORM) to cut manual work and errors and enable data-driven decisions.

Problem Statement & Objectives

- Problem:** Manual order processing, inventory issues, delayed deliveries, hard-to-track returns and poor insight into customer buying behavior.
- Objective 1:** Manage the full order lifecycle: placement, processing, shipping, delivery, cancellation and returns.
- Objective 2:** Monitor inventory with low-stock alerts and improve retention via notifications and loyalty points.
- Objective 3:** Deliver analytics and decision-support insights from sales, inventory and return data.
- Scope:** Admin, Staff and Customer roles across 5 modules: Access & Roles, Core E-Commerce, Engagement & Loyalty, Analytics, Decision Support.

Existing vs Proposed System

Feature	Existing System	Proposed System
Order Handling	Manual, error-prone processing	Automated order lifecycle with status tracking
Inventory	Stock issues, no alerts	Live stock monitoring + low-stock alerts
Returns	Difficult to track	Structured return & refund workflow
Customers	No history or rewards	Profiles, notifications, loyalty points
Limitation / Gap	Limited view of sales & buying behavior	Chart.js dashboard + decision-support insights

Feasibility Study

- Technical:** Team skilled in React.js, Node.js/Express.js and MySQL; all tools (VS Code, Git, Node.js v18+, MySQL 8.0+) are freely available.
- Economic:** Fully open-source stack with no licensing cost; deployable on free-tier hosting.
- Operational:** Simple role-based interface for admins, staff and customers; modular code is easy to maintain.
- Schedule:** Five modules split among three members, achievable within the semester timeline.

Results & Testing

- Core flows tested: product, order, inventory and return CRUD.
- Validation and JWT role-based access checks on API routes.
- Dashboard charts and insights built from live MySQL data.

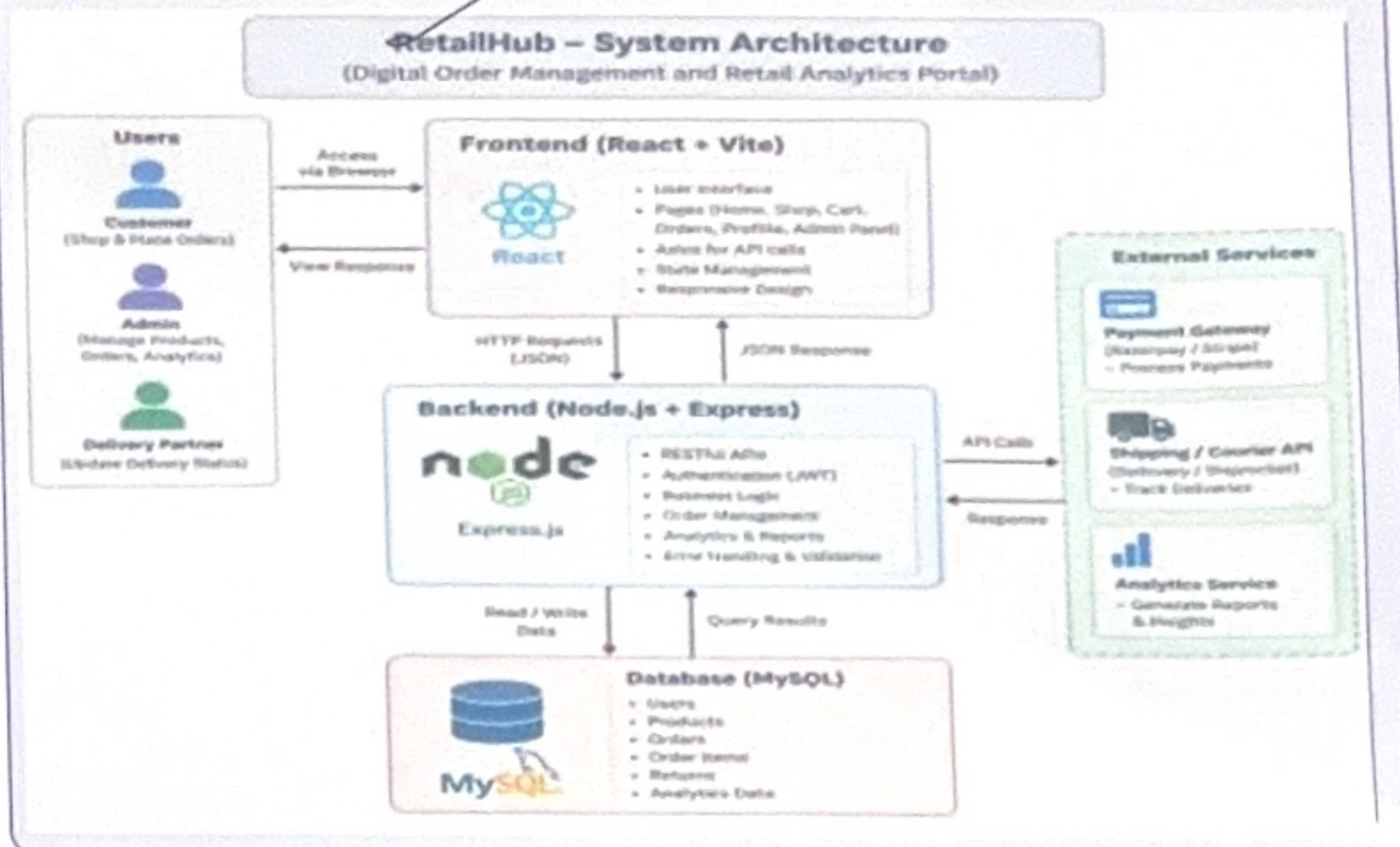
Conclusion

- One platform unifying order management, inventory control, customer engagement and analytics.
- Meets all objectives: fewer manual errors, better efficiency, data-driven decisions and happier customers.

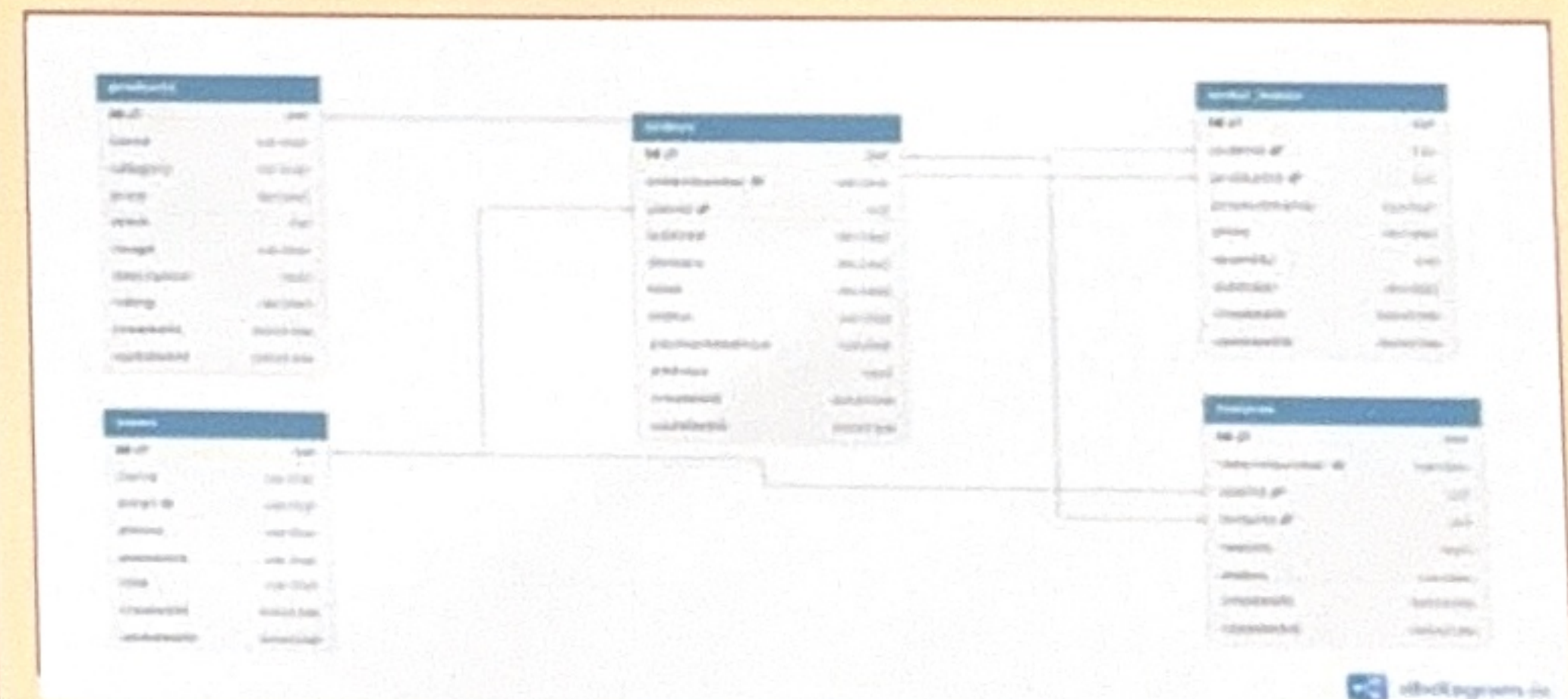
Tools & Technologies

- Front-End:** React.js, Chart.js
- Back-End:** Node.js v18+, Express.js, Sequelize ORM, JWT + REST APIs
- Database:** MySQL Server v8.0+
- Tools:** Visual Studio Code, Git

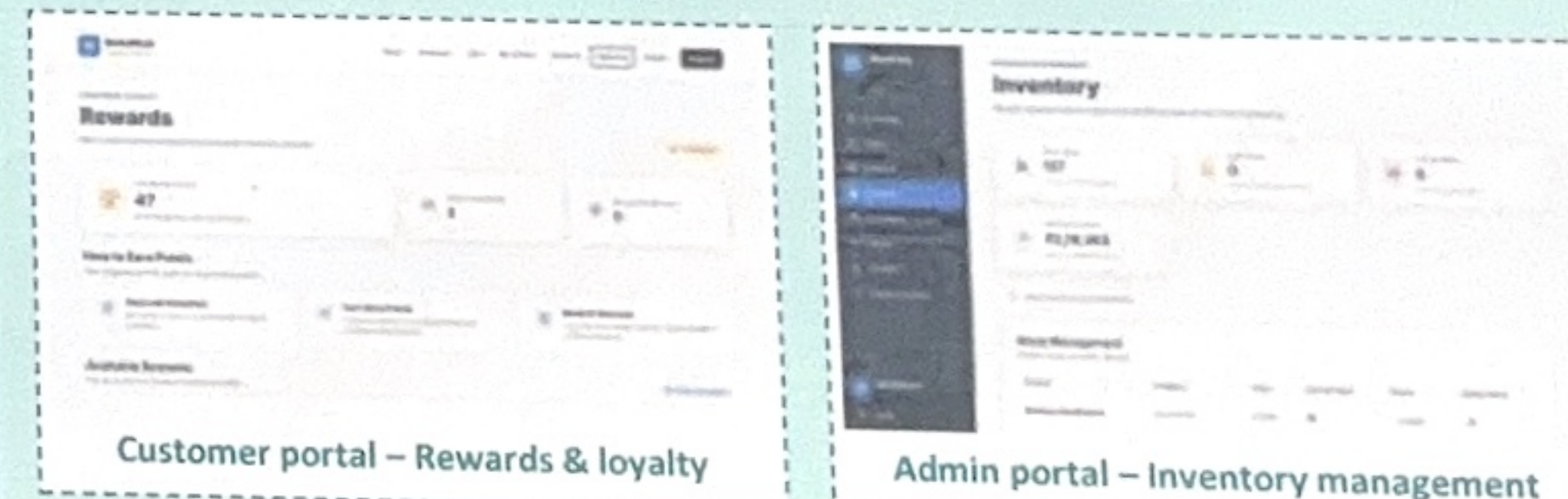
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

- Payment gateway and delivery-partner tracking integration.
- Demand forecasting and personalized recommendations.
- Mobile app with email/SMS notifications.